YOUNGJIN KIM

Postdoctoral Research Fellow

Stanford University, United States

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EDUCATION

Ph.D.	**Electrical and Computer Engineering** Seoul National University, South Korea Advisor: Byoungho Lee (deceased) Yoonchan Jeong	2019 – 2024
B.S.	**Electrical and Computer Engineering** Seoul National University, South Korea Advisor: Byung Gook Park (deceased)	2012 – 2019

CAREER

Postdoc	**Stanford University**, CA, USA Advisor: Mark Brongersma	2025 –
Postdoc	**Seoul National University**, South Korea Advisor: Yoonchan Jeong	2025
Intern	**Meta Reality Labs**, WA, USA Department: Optics, Photonics, and Light Systems (OPALS) Research Research subject: Development of new waveguide architecture Manager: Xiayu Feng	2024

HONORS AND AWARDS

- **Doyeon Academic Paper Award**
  Inter-university Semiconductor Research Center, Seoul National University, 2025
- **Sejong Science Fellowship (Overseas Training Track)**
  National Research Foundation of Korea, 2025.
- **Distinguished Dissertation Award**
  Seoul National University, 2025.
- **Seoul National University Joint International Research Grant**
  Seoul National University, 2023.
- **Silver Prize, Samsung Display Industry-Academia Technical Paper Awards**
  Samsung Display, 2023.
- **Scholarship of Foundation for SNU ECE - Kim Jung Sik Fund**
  Seoul National University, 2021.

- **Best Poster Paper Awards**
  Optics and Photonics Congress, Jeju, South Korea, 2021.

JOURNAL PUBLICATIONS

First Author (4)

† : equal contribution

1. **Y. Kim**†, T. Choi†, G.-Y. Lee, C. Kim, J. Bang, J. Jang, Y. Jeong, and B. Lee, "Metasurface Folded Lens System for Ultrathin Cameras," **Science Advances**, vol. 10, no. 44, pp. eadr2319, 2024.
2. **Y. Kim**, G.-Y. Lee, J. Sung, J. Jang, and B. Lee, "Spiral Metalens for Phase Contrast Imaging," **Advanced Functional Materials**, vol. 32, no. 5, pp. 2106050, 2022.
3. S.-W. Nam†, **Y. Kim**†, D. Kim, and Y. Jeong, "Depolarized Holography with Polarization-multiplexing Metasurface," **ACM Transactions on Graphics (SIGGRAPH Asia),** vol. 42, no. 6, article 200, 2023.
4. Y. Park†, **Y. Kim†,** C. Kim, G.-Y. Lee, H. Choi, T. Choi, Y. Jeong, and B. Lee, "End-to-end Optimization of Metalens for Broadband and Wide-angle Imaging," **Advanced Optical Materials**, vol. 13, no. 9, pp. 2402853, 2025.

Co-Author (8)

1. G.-Y. Lee†, C. Kim†, M. Gopakumar, **Y. Kim**, B. Lee, Y. Jeong, and G. Wetzstein, "Neural phase microscopy with metasurface optics for real-time and nanoscale quantitative phase imaging," **Nature Communications**, vol. 17, no. 1411, 2026.
2. J. Bang, **Y. Kim**, T. Choi, C. Kim, H. Son, S.-J. Kim, Y. Jeong, and B. Lee, "Cascaded Janus meta-optics: generalized platform for bidirectional asymmetric modulation of light," **ACS Photonics**, vol. 12, no. 3, pp. 1666-1675, 2025.
3. C. Kim, J. Hong, J. Jang, G.-Y. Lee, **Y. Kim**, Y. Jeong, and B. Lee, "Freeform Metasurface Color Router for Deep Sub-micron Pixel Image Sensors," **Science Advances**, vol. 10, no. 22, pp. eadn9000, 2024.
4. T. Choi, C. Choi, J. Bang, **Y. Kim**, H. Son, C. Kim, J. Jang, Y. Jeong, and B. Lee, "Multiwavelength Achromatic Deflector in the Visible Using a Single-Layer Freeform Metasurface," **Nano Letters**, vol. 24, no. 35, pp. 10980-10986, 2024.
5. H. Son, T. Choi, K. Kim, **Y. Kim**, J. Bang, S.-J. Kim, B. Lee, and Y. Jeong, "Strong Coupling Induced Bound States in the Continuum in a Hybrid Metal–Dielectric Bilayer Nanograting Resonator," **ACS Photonics**, vol. 11, no. 8, pp. 3221-3232, 2024.
6. E. Lee, Y. Jo, S.-W. Nam, M. Chae, C. Chun, **Y. Kim**, Y. Jeong, and B. Lee, "Speckle Reduced Holographic Display System with a Jointly Optimized Rotating Phase Mask," **Optics Letters**, vol. 49, no. 19, pp. 5659-5662, 2024.
7. J. Jang, G.-Y. Lee, **Y. Kim**, C. Kim, Y. Jeong, and B. Lee, "Dispersion-Engineered Metasurface Doublet Design for Broadband and Wide-Angle Operation in the Visible Range," **IEEE Photonics Journal**, vol. 15, no. 4, pp. 1-9, 2023
8. S.-J. Kim, C. Kim, **Y. Kim**, J. Jeong, S. Choi, W. Han, J. Kim, and B. Lee, "Dielectric metalens: properties and three-dimensional imaging applications," **Sensors**, vol. 21, no. 13, pp. 4584, 2021.

PATENTS

1. Y. Park, **Y. Kim**, G.-Y. Lee, B. Lee, Y. Jeong, "Double sided metalens and electronic device including the same" (US – Application No. 18/490,121)
2. Y. Jeong, S.-W. Nam, **Y. Kim**, D. Kim, "Holographic display using metasurface and metasurface optimization method" (US – Application No. 18/737,648)
3. J. Hong, C. Kim, B. Lee, **Y. Kim**, G.-Y. Lee, J. Jang, Y. Jeong, "Color-routing element, method of manufacturing the same, and image sensor including the color-routing element" (US – Application No. 18/937,958)

INVITED TALKS

Mar 2026 **University College London (UCL)**, invited by Prof. Kaan Akşit
Nanophotonics for Next-Generation Imaging Systems

Feb 2024 **Stanford University**, invited by Prof. Mark Brongersma
Metasurface optics for imaging and holographic display systems

SERVICES

Reviewer

Nature Communications, Scientific Reports, ACS Photonics, ACM SIGGRAPH, ACM SIGRGAPH Asia, ACM Transactions on Graphics (TOG), Optical and Quantum Electronics